Lachlan Watts-Tobin

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EDUCATION

University of California, Berkeley Berkeley, CA
B.S. Electrical Engineering & Computer Sciences (GPA: 3.3 / 4.0) May 2028
Coursework: Digital Design and Integrated Circuits (with ASIC Lab), Computer Architecture, Internet of Things, Machine Structures, Circuits and Devices, Data Structures, Signals and Information Processing, Solid Mechanics.

SKILLS

Languages: SystemVerilog, Verilog, C, C++, CUDA, Python, Chisel, Scala, RISC-V, TCL, Perl, MATLAB.
Tools: Cadence (Innovus, Indago), Synopsys (VCS, Verdi), UVM, Linux, Git, PyTorch, TensorRT, Pandas, NumPy.

EXPERIENCE

Pano AI San Francisco, CA
Hardware Engineering Intern May 2026 – August 2026
- Designed an edge-computer system layout optimizing area efficiency and minimizing power leakage.
- Authored Manufacturing Process Information (MPI) to scale edge-computer production.
- Developed a 94%-accurate lens-contamination detection algorithm flagging water/debris to preserve quality.

Vision IV (Remote) Berkeley, CA
Hardware Engineering Intern May 2025 – July 2025
- Optimized a high-performance capacitor electrode using experimental materials to improve potential difference.
- Analyzed failure points and ran reliability testing to validate the design's performance.

UC Berkeley TAF Lab Berkeley, CA
Undergraduate Researcher Sept 2026 – Present
- Design, bring up, and optimize microcontrollers for a swarm of autonomous sailboats (USVs) for ocean sensing.
- Conduct on-site field testing and live debugging of boat electronics during on-water deployments.

PROJECTS

SkyWater 130nm 5-Stage Pipelined SoC *October 2026 – Present*
- Designing a 5-stage pipelined core with I/D caches and branch prediction in the SkyWater 130nm process.
- Implementing the RTL datapath and control in SystemVerilog, utilizing the Hammer VLSI flow.
- Verifying the core with functional tests and UVM while iterating on timing and area closure toward tapeout.

Logisim Pipelined CPU Core *November 2025 – December 2025*
- Designed a 5-stage pipelined CPU in Logisim executing a 32-bit RISC-V instruction subset.
- Resolved data and control hazards with forwarding and stalling to keep dependent instructions correct.
- Validated the datapath and control against directed RISC-V test programs across all pipeline stages.

Scheme Language Interpreter *March 2025 – April 2025*
- Implemented an interpreter for a subset of Lisp (Scheme), supporting core syntax, macros, and expressions.
- Built the read–eval loop with lexical analysis, parsing, and evaluation, including recursion and tail calls.

AI-Powered Guitar Pedal *June 20, 2026*
- Bridged AI to hardware, pushing generated parameters from plain English to an ESP32 DSP device over WiFi.

Automated Trading Card Sorter *April 2026 – May 2026*
- Engineered a 98%-accurate machine that sorts trading cards by their live value, using CV, APIs and web scraping.

Self-Hosted Portfolio Website *May 2026 – August 2026*
- Self-hosted a portfolio site on a Raspberry Pi via a tunnel with custom DNS and automated Git deploys.

EXTRACURRICULARS

Bike Builders of Berkeley Berkeley, CA
Lead Electrical Engineer August 2024 – Present
- Lead the electrical design of the e-bike's EV drivetrain, including custom PCBs and controllers.
- Balance workloads across website, firmware, and hardware as project manager.

UC Berkeley, College of Engineering Berkeley, CA
Undergraduate Tutor July 2025 – Present
- Tutor lower-division electrical engineering and computer science coursework to students every weekday.